

Concept Down-Selection Review

Team Identity



Executive Summary

This report demonstrates the concept down selection for the 2040 cargo aircraft which was focused on reducing the direct operating cost & climate impact by focusing on regulatory and operational constraints. To assess candidate configuration, a multi criteria decision matrix and further analysis which were driven by customer priorities and then translating them into engineering weightings using a “House of Quality” approach was followed. The conventional configuration with wide body and folding wing tips has been chosen for further development because it gives the maximum fuel efficiency and to ensure technological readiness.

Team Roles

The table below shows a summary of the team members and respective roles.

Role Name	Team Member
Project Manager / Economics Engineer	Mapalo
Wing Structures Engineer	Nithin
Fuselage Structures Engineer	Santosh
Wing Aerodynamics Engineer	Nethaji
Empennage & Stability Engineer	Oscar
Performance, Propulsion & Climate	Henri-Alexandre de Croy-Solre
Optimisation & Integration Engineer	Kamran
Landing Gear Engineer	N/A

Fleet Level Top-Level Aircraft Requirements

Fleet Level

TLAR	Value	Unit (SI)
Max Range	10186	km
Max number of stops	1	N/a
ETOPS Time Limit	180	min
Minimum Time to Climb 457.2 m to 6096 m	30	min
Minimum cruise altitude	6096	m
Total Payload	736	tonnes

Aircraft Level

TLAR	Value	Unit (SI)
Wingspan Airport Compatibility with Code	F & E	N/a
Airworthiness Certification	CS-25/FAR-25	N/a
Minimum take-off/landing distance	2900	m
Maximum Pallet capability	P6P+	N/a
Fuel type	Kerosene or SAF	N/a
Cabin Pressure at cruise altitude equivalent to	2438.4	m
Minimum Mach Number	0.502	N/a

The aircraft and fleet as a whole is constrained through geometric, structural, aerodynamic, performance and certification/regulation limits. Taken together, these requirements determine the factors that ultimately drive both direct operating cost and climate impact.

Top Level Aircraft Requirements (TLARs): Airport code (E/F) compatibility and P6P+ pallet capability directly define fuselage volume and wingspan, which in turn determine cargo layout, fuel volume and aerodynamic efficiency. Take-off/landing distance, minimum Mach number, fuel type and cabin pressurisation requirements force having to trade-offs between engine thrust/SFC, structural weight and tank capacity. Compliance with CS-25/FAR-25 and associated reliability requirements imposes redundancy that increases overall aircraft mass. Taken together, these constraints define what the aircraft can physically be.

Top Level Fleet Requirements (TLFRs): Maximum range, ETOPS time limit, maximum number of stops, minimum cruise altitude, total payload and time-to-climb define routing flexibility, mission tempo as well as overall fleet output. These parameters determine how effectively the fleet can move the required cargo.

Concept Down Selection Methodology

This section outlines the process by which the final two concepts have been selected, to be analysed through the MDO development sprint.

Customer Criteria

The customer criteria selected below reflect the key factors influencing aircraft choice from an operator's perspective. Collectively, these requirements ensure that the final aircraft selection achieves a balanced trade-off between cost, operational performance, environmental responsibility, and technical risk.

Criteria	Weighting	Description
Direct Operating Cost (DOC)	0.29	Represents the overall cost of operating the aircraft, including energy, maintenance, crew, and ownership costs, and directly affects airline profitability.
Climate Impact / Emissions	0.23	Considers the aircraft's environmental footprint, including emissions, noise, and its ability to meet current and future environmental regulations. → directly related to engines

Turnaround Time / Ease of Loading	0.14	Evaluates how quickly the aircraft can be unloaded, serviced, refuelled, and prepared for the next flight, directly affecting aircraft utilisation and schedule reliability.
Reliability/ Maintenance	0.15	Reflects expected operational reliability, standard maintenance requirements/ downtime, affecting service continuity and costs.
Development/Operational Risk	0.13	Evaluates how well-proven the technologies are, influencing certification risk, insurance costs, and operator confidence.
Schedule flexibility	0.07	Indicates the aircraft's ability to handle route deviations or diversions without significant performance or range penalties.

Engineering Criteria

Meeting the specification requirements requires balancing several competing engineering factors, including payload capability, range, cruise performance, and overall efficiency, all within strict MTOM limits. Similarly, meeting the target cruise Mach number while maintaining acceptable fuel consumption introduces additional aerodynamic and engine integration challenges. These performance targets must be achieved without relying on technologies that are unlikely to be mature or certifiable by the 2040 entry-into-service date. To guarantee the concept's industrial viability, manufacturing and development costs must be managed concurrently.

To compare the proposed concepts fairly, a set of independent evaluation criteria was selected, each representing a different aspect of performance, feasibility, or cost. The chosen criteria and their relative weightings are shown in Table below, with the weighting selection process described in the following section.

Criteria	Weighting	Brief Description
Technology Readiness Level (TRL)	0.134	Maturity and certification readiness of applied Technologies (2040)
Maximum Payload mass	0.157	Ability to carry the required maximum payload within MTOM limits.
Cruise Mach Number	0.135	Ability to meet required cruise Mach number efficiently.

Cruise Lift to drag ratio(L/D)	0.129	Aerodynamic efficiency of the aircraft in cruise, influencing fuel consumption, range, and overall performance.
SFC of Engines (Kg)	0.176	Total Fuel Consumed over the design mission.
Range at Max Payload + MTOW	0.082	Ability to meet the fixed design range at maximum payload.
Manufacturing + Development Cost	0.187	Relative cost and industrial complexity of the concept.

Criteria weightings

		Cruise Mach number (+)	Cruise lift-to-drag ratio (L/D) (+)	Maximum payload mass (+)	Technology readiness level (TRL) (+)	SFC of Engines (kg)	Range at Max Payload + MTOW (km)	Manufacturing /Development Costs	Favourability
Person	Weight	↑	↑	↑	↑	↓	↑	↓	
Direct Operating Cost (DOC)	0.29	5	6	4	2	7	3	8	
Climate Impact / Emissions	0.23	4	6.5	7	3	9	2	6	
Turnaround Time / Ease of loading	0.14	0	0	7	2	0	0	2	
Reliability / Maintenance	0.15	6.5	1	5	8	2.5	1.5	6	
Development /Operational Risk	0.13	2.5	3	2	10	5	3.5	7	
Schedule flexibility	0.07	8	4	3	3	6	8	2	Sum
	Score	4.22	4.04	4.91	4.19	5.52	2.57	5.86	31.33
	Weighting	0.135	0.129	0.157	0.134	0.176	0.082	0.187	1.000

A simplified “House of Quality (HoQ)” method was used to translate customer & engineering priorities into as objective as possible weights! The HoQ matrix was populated by allowing every team member to pick an estimated value for the criteria (under review). To try and eliminate bias as much as possible, every team member was asked to refer back to the original specification as a primary source of information to try and provide the best possible assumption of the relevance parameter under review. As

an example, in the final table Direct Operating Cost (DOC) and emissions map strongly to SFC and L/D, while development/insurance risk was almost solely rated based off of the technology readiness level. Where an engineering criterion does not meaningfully affect a customer requirement, the relationship score is set low or zero (i.e SFC of the engines has no direct impact on the turnaround time).

Favourability is indicated by “↑/↓” arrows as to whether we are maximising or minimising the intended outcome of the variable (e.g., lower SFC and lower development cost are favourable; higher L/D, payload, and range are favourable). This provides clear traceability as to the intended outcome of providing a certain value in the matrix (from customer perspective).

Concept Down Selection

Multi-Criteria Decision Analysis (MCDA)

Multi-Criteria Decision Analysis (MDCA)		Conventional	Strut-Based	Wide Conventional + folding wing tips	Double fuselage - Folding wing tips
Criteria	Weight	CON1	CON2	CON3	CON4
Cruise Mach number (+)	0.135	1	-1	1	0
Cruise lift-to-drag ratio (L/D) (+)	0.129	1	4	2	1
Maximum payload mass (+)	0.157	1	-1	4	1
Technology readiness level (TRL) (+)	0.134	1	-1	1	-2
SFC of Engines (kg)	0.176	1	2	-2	-1
Range at Max Payload + MTOW (km)	0.082	1	2	-2	0
Manufacturing + Development Costs	0.187	1	-2	-1	-2
		1	0.23	0.45	-0.53

Alternative configuration

The MCDA comparison explores the trade-off between performance gains (cruise speed, aerodynamic efficiency, payload and mission fuel) and deliverability (technology

maturity and programme cost) across concepts (CON1 – CON4 where CON1 is conventional). The goal of this analysis is to identify the concept which has the overall best potential for our mission criteria, primarily weighted by Manufacturing + Development Costs (0.187) and Fuel burn (0.176), followed by Maximum payload (0.157), then Cruise Mach (0.135). TRL (0.134) was also a critical factor that had to be considering, knowing that the brief specifies that the design must be ready for 2040. This tight timeline on the aircraft development weighted heavily against designs like the truss braced wing due to it's lack of use in the transport industry, or in any use case at all.

The selected alternative configuration is the **Wide Conventional + Folding Tips (CON3)**. It retains a conventional tube-and-wing architecture but with a wider fuselage. The folding wing tips as an idea, came about because of the gate capacity for the runway and taxiing (code E and F) given by the brief to ensure that that the plane is capable of fitting in most standard airports. The key design elements driving selection are therefore increased payload capability and high programme maturity (Looking at individual aircraft such as the Antonov 124 and 225 as well as military aircraft like the Lockheed C5 supergalaxy), while avoiding the major integration and certification risks associated with more disruptive layouts like the BWB or additional truss on the wing. CON3 was scored +4 for maximum payload (weight 0.157) because the widened fuselage capacity directly supporting higher payload volume versus the baseline CON1. It was scored +1 for Cruise Mach (0.135) because the group plans to preserve standard swept wing high subsonic cruise characteristics, preparing for inevitable transonic flight. It was also scored +1 for TRL (0.134) as it remains within established/existing designs (like the AN-124 mentioned above), but no more arguably so than a conventional aircraft like the current solution B777F.

Some of the negatives of this design to consider are CON3 received –2 for fuel burn (0.176) (Expecting the plane to be less efficient even despite the larger aspect ratio – this logic however will potentially not apply to the whole fleet as on later analysis we may find we don't need as many aircraft. The cost score of –1 (0.187) is due to the fact that some specific development may be required into the wider body when handling the uneven pressure distribution on the body (this isn't necessarily a widely seen concept as a whole) also obviously extra materials are required however this is minimal compared to the research. As a result, it is still far less programme risky than unconventional designs. Overall, the weighted balance favours CON3 as the most deliverable improvement aligned with the criteria priorities provided by the specification.

Hyper-parameters

The following table highlights the hyperparameters chosen for the optimisation process. The range of each parameter was selected by analysing the specifications of cargo aircraft, both conceptual and commercial aircraft, varying in size, range and payload capacity, and considering how these existing aircraft will be utilised for this application.

Name	Range	Unit
Number of Aircraft	5-30	-
Wing Aspect Ratio	7.8 - 11	-
Payload Capacity	30000-150000	kg
Range at Maximum Payload	900-5500	km
Maximum Take-Off Weight	187000 – 575000	kg
Sweep Angle	25-40	degrees
Foldable Wing Tip Length	3-4	m
Payload Volume	500 - 1900	m ³
Cruise Mach Number	0.502-0.92	Mach